Optimal Dynamic Information Acquisition

Presentation Notes on Zhong (2022, *Econometrica*)
with insights from Sannikov & Zhong (2024)

Presentation Notes

Abstract

These notes prepare a 25-minute presentation of the main results in Zhong (2022). The audience is an information design class familiar with Bayesian persuasion, concavification, and static rational inattention, but not necessarily comfortable with continuous-time stochastic calculus. The notes explain every symbol carefully and develop intuition before formalism.

Contents

1 Part 1: The Problem and the Punchline (5 min) 2
1.1 The question in words . . . 2
1.2 Primitives . . . 2
1.3 Beliefs as a stochastic process . . . 2
1.4 The cost of learning . . . 3
1.5 The DM's optimisation problem . . . 3
1.6 The punchline . . . 4

2 Part 2: The Main Characterisation (10 min) 5
2.1 Reducing the problem: from all martingales to Poisson + Gaussian . . . 5
2.2 The Hamilton–Jacobi–Bellman (HJB) equation . . . 5
2.3 The main characterisation theorem (binary state) . . . 7
2.4 The dynamics in words . . . 7

3 Part 3: Why? The Key Tradeoff (8 min) 9
3.1 The precision–frequency tradeoff . . . 9
3.2 The gross value function and concavification . . . 9
3.3 How the tradeoff resolves: varying λ . . . 10
3.4 Why confirmatory? (Property (ii)) . . . 10
3.5 Why no repeated jumps? (Property (v)) . . . 10
3.6 Why not Gaussian? (Property (i)) . . . 11
3.7 Where Zhong (2022) sits in the bigger picture . . . 11

1 Part 1: The Problem and the Punchline (5 min)

1.1 The question in words

A decision-maker (DM) must choose an action but does not know the state of the world. Before acting, she can **learn**—she can design experiments, collect data, run tests. Learning takes time (the DM discounts future payoffs), and learning is costly (more informative experiments cost more per unit of time).

The central question is: **if you can flexibly design any experiment you like, at each instant in time, what does the optimal learning strategy look like?**

The answer is striking: the optimal strategy resembles a **Poisson process**. The DM waits for a rare but decisive "breakthrough." If the breakthrough arrives, she acts immediately. If it does not arrive, she keeps waiting, but her experiments become increasingly stringent (harder to pass) and less frequent over time.

1.2 Primitives

There is an unknown state of the world x, drawn from a finite set X. The DM holds a **prior belief** $\mu_0 \in \Delta(X)$.

Notation: $\Delta(X)$

$\Delta(X)$ is the set of all probability distributions over X. When $X = \{0, 1\}$ (two states), we can identify $\mu \in \Delta(X)$ with a single number $\mu \in [0, 1]$: the probability assigned to state 1.

The DM must choose an action a from a finite set A. The payoff from action-state pair (a, x) is $u(a, x)$. If the DM knew the state, she would pick the best action. Since she only has belief μ, the best she can do *right now* is:

$$F(\mu) \triangleq \max_{a \in A} \mathbb{E}_\mu\big[u(a, x)\big] = \max_{a \in A} \sum_{x \in X} \mu(x)\, u(a, x).$$

Why F is convex

$F(\mu)$ is the maximum of a collection of linear functions of μ (one for each action a). The maximum of linear functions is convex. This convexity is the source of the *value of information*: by learning (spreading out the belief), the DM can increase expected payoffs—this is exactly the logic behind Bayesian persuasion.

1.3 Beliefs as a stochastic process

Rather than specifying signal structures (signal spaces, likelihood functions), Zhong models the DM as directly choosing the **belief process** $\langle \mu_t \rangle_{t \geq 0}$.

Time is continuous: $t \in [0, +\infty)$. At every instant t, the DM's belief is $\mu_t \in \Delta(X)$.

The key constraint is Bayes' rule, which requires the belief process to be a **martingale**:

$$\mathbb{E}[\mu_s \mid \mathcal{F}_t] = \mu_t, \qquad \forall\, s > t.$$

What this means

$\mathcal{F}_t$ is the "information available at time t"—everything the DM has observed up to time t. The martingale condition says: *my best forecast of my future belief is my current belief.* This is not an assumption about rationality; it is a logical consequence of Bayes' rule. On average, I don't expect my belief to go up or down—I just don't know *which way* it will move.

Equivalently, if at time t I expected my time-s belief to be higher than my current belief, I should update now. The martingale property rules out such foreseeable revisions.

1.4 The cost of learning

Learning is measured by an **uncertainty function** $H : \Delta(X) \to \mathbb{R}$, which is strictly concave. Think of $H(\mu)$ as quantifying "how uncertain belief μ is." For instance:

- Shannon entropy: $H(\mu) = -\sum_x \mu(x) \log \mu(x)$.
- Quadratic: $H(\mu) = \mu(1-\mu)$ (in the binary case).

At each instant, the DM acquires information at a rate measured by how fast uncertainty is falling:

$$I_t \;=\; -\lim_{s\to 0} \frac{\mathbb{E}\big[H(\mu_{t+s}) - H(\mu_t) \,\big|\, \mathcal{F}_t\big]}{s} \;\geq\; 0.$$

Why $I_t \geq 0$

H is concave and μ_{t+s} is a mean-preserving spread of μ_t (by the martingale property). Jensen's inequality gives $\mathbb{E}[H(\mu_{t+s}) \mid \mathcal{F}_t] \leq H(\mu_t)$. So the numerator is non-positive, and $I_t \geq 0$. Learning always reduces expected uncertainty. The rate of this reduction is I_t.

The flow cost of information is $C(I_t)$, where $C : \mathbb{R}^+ \to \mathbb{R}^+$ is:

- **Convex**: increasing marginal cost of learning faster.
- $C'(\infty) = \infty$: the DM cannot learn arbitrarily fast.
- $C(0) = 0$: not learning is free.

1.5 The DM's optimisation problem

The DM discounts the future at rate $\rho > 0$. She chooses a belief process $\langle \mu_t \rangle$ (any martingale) and a **stopping time** τ (the moment she acts) to maximise:

$$V(\mu) \;=\; \sup_{\langle \mu_t \rangle,\, \tau} \mathbb{E}\left[e^{-\rho\tau} F(\mu_\tau) \;-\; \int_0^\tau e^{-\rho t} C(I_t)\, dt \right]. \tag{1}$$

Reading equation (1)

- $e^{-\rho\tau} F(\mu_\tau)$: the payoff from acting at time τ with belief μ_τ, discounted.
- $\int_0^\tau e^{-\rho t} C(I_t)\, dt$: the total accumulated cost of learning, also discounted.
- The DM chooses *both* the process (what to learn) and the stopping time (when to stop).
- $V(\mu)$ is the *value function*: the maximum expected payoff when starting from belief μ.

1.6 The punchline

Main result (Theorem 2, informal)

The optimal belief process is a **compensated Poisson process**. At each instant:

1. With small probability $p(\mu_t)\,dt$, the belief **jumps** to a target $\nu(\mu_t)$ and the DM **stops immediately**. This is the "breakthrough."
2. With probability $1 - p(\mu_t)\,dt$, nothing happens, and the belief **drifts** smoothly in the opposite direction (by Bayes' rule, the absence of the breakthrough is itself informative).

Moreover, the breakthrough **confirms** the prior (pushes the belief toward the state the DM already finds most likely), and over time, breakthroughs become **rarer but more decisive**.

Why Poisson and not Gaussian?

Sannikov & Zhong (2024, Proposition 2) give a one-line conceptual explanation. Exponential discounting means the "utility of stopping at time t" is $e^{-\rho t}F(\mu)$, which is **convex in** t (because $e^{-\rho t}$ is convex).

By Jensen's inequality, a convex function of a random variable favours **more dispersed** distributions. The DM is effectively **time-risk loving**: she prefers a lottery where she either stops very quickly or waits a long time, over one where she always stops at the average time.

Poisson learning maximises this dispersion—the breakthrough either comes very soon (with some probability) or takes a very long time (the remaining probability). Gaussian learning, by contrast, produces a "hump-shaped" distribution of stopping times with much less dispersion. So exponential discounting pushes toward Poisson.

2 Part 2: The Main Characterisation (10 min)

2.1 Reducing the problem: from all martingales to Poisson + Gaussian

The DM's control variable is an entire stochastic process $\langle \mu_t \rangle$—an infinite-dimensional object. The first result drastically simplifies this.

The idea is as follows. At each instant t, the DM chooses the *conditional distribution* of μ_{t+dt} given $\mu_t = \mu$. Call this distribution π. The martingale constraint requires $\mathbb{E}_\pi[\nu] = \mu$ (the mean of the next-period belief equals the current belief).

The cost of information at this instant is $C\Big(\frac{1}{dt}\,\mathbb{E}_\pi\big[H(\mu) - H(\nu)\big]\Big) \cdot dt$: the cost depends on the rate at which uncertainty falls.

The key observation is that both the *benefit* $\frac{1}{dt}\mathbb{E}_\pi[V(\nu) - V(\mu)]$ and the *information* $\frac{1}{dt}\mathbb{E}_\pi[H(\mu) - H(\nu)]$ are *linear* in π. So for a given information level I, the problem of choosing π to maximise the benefit is a **linear program**. A linear program over probability measures with finitely many constraints has a solution with **finite support** (Carathéodory's theorem).

As $dt \to 0$, the support points of π that stay near μ merge into a **diffusion** (Gaussian component); at most one support point stays finitely far from μ and becomes a **Poisson jump**. The result is:
Theorem 1 (Theorem 1 of Zhong (2022), simplified)**.** *It suffices to consider belief processes of the form:*

$$d\mu_t = \underbrace{(\nu(\mu_t) - \mu_t)\big(dJ_t - p(\mu_t)\,dt\big)}_{\textit{compensated Poisson}} + \underbrace{\sigma(\mu_t)\,dW_t}_{\textit{Gaussian diffusion}}, \tag{2}$$

where J_t is a Poisson counting process with arrival rate $p(\mu_t)$, and W_t is a standard Brownian motion.

Reading equation (2)

This equation describes how the belief μ_t evolves over an infinitesimal time interval dt. There are two parts:

Poisson part. J_t is a counting process: it stays at 0 for a random amount of time, then jumps to 1, stays at 1, jumps to 2, and so on. The time between consecutive jumps is exponentially distributed with rate $p(\mu_t)$.

- *When J_t jumps* (probability $\approx p\,dt$ per instant): the belief jumps from μ_t to $\nu(\mu_t)$. This is the "breakthrough." The term $(\nu - \mu_t) \cdot (dJ_t)$ captures this jump.
- *When J_t does not jump* (probability $\approx 1 - p\,dt$): the "$-p\,dt$" term creates a deterministic drift of $-p(\mu_t)\big(\nu(\mu_t) - \mu_t\big)\,dt$. This drift compensates for the jump, so that on average the process doesn't move—preserving the martingale property.

Gaussian part. W_t is a standard Brownian motion—a continuous random process where dW_t is a normal random variable with mean 0 and variance dt. So $\sigma(\mu_t)\,dW_t$ adds a continuous, normally distributed perturbation with standard deviation $\sigma\sqrt{dt}$ per instant.

The controls are the functions $p(\cdot)$ (the Poisson rate), $\nu(\cdot)$ (the jump target), and $\sigma(\cdot)$ (the diffusion volatility). Theorem 1 says these three functions are all you need to optimise over—you do not need to consider more exotic processes.

2.2 The Hamilton–Jacobi–Bellman (HJB) equation

Given that the belief process has the form (2), the value function $V(\mu)$ satisfies a differential equation. At each instant, the DM either stops (gets $F(\mu)$) or continues. If she continues, her

value $V(\mu_t)$ changes due to the movement of the belief. The "flow change" in V is captured by a quantity called the **generator** $\mathcal{A}V$:

$$\mathcal{A}V(\mu) \;=\; \underbrace{p\big(V(\nu) - V(\mu) - V'(\mu)(\nu-\mu)\big)}_{\text{Poisson contribution}} \;+\; \underbrace{\tfrac{1}{2}\,\sigma^2\,V''(\mu)}_{\text{Gaussian contribution}} \quad . \tag{3}$$

Reading the generator (3)

For the binary case, $\mu \in [0,1]$ is a scalar, so V' and V'' are ordinary derivatives.

Poisson term:

- $V(\nu) - V(\mu)$: the change in value if the belief jumps from μ to ν.
- $-V'(\mu)(\nu-\mu)$: subtracting the first-order approximation. What remains is the "surprise" component of the jump—the part not captured by the local slope.
- $p(\cdots)$: multiply by the jump rate because the jump happens with probability $p\,dt$ per instant.

Geometrically, $V(\nu) - V(\mu) - V'(\mu)(\nu-\mu)$ is the gap between $V(\nu)$ and the tangent line of V at μ, evaluated at ν. When V is convex, this gap is positive: the Poisson jump *increases* value.

Gaussian term:

- $V''(\mu)$: the curvature (second derivative) of V.
- $\frac{1}{2}\sigma^2$: half the instantaneous variance of the diffusion.

When V is convex ($V'' > 0$), the Gaussian noise also *increases* value (by Jensen's inequality applied to a convex function).

Similarly, the flow information measure is:

$$I \;=\; \underbrace{p\big(H(\mu) - H(\nu) + H'(\mu)(\nu-\mu)\big)}_{\text{information from Poisson signal}} \;+\; \underbrace{\big(-\tfrac{1}{2}\,\sigma^2\,H''(\mu)\big)}_{\text{information from Gaussian signal}} \quad . \tag{4}$$

Reading the information measure (4)

This has the same structure as the generator but with H (concave) in place of V (convex), and with a sign flip.

- $H(\mu) - H(\nu) + H'(\mu)(\nu-\mu)$: the gap between the tangent line of H at μ and $H(\nu)$. Since H is concave, the tangent line is *above* $H(\nu)$, so this is positive. This is the **Bregman divergence** $D_H(\nu\|\mu)$.
- $-\frac{1}{2}\sigma^2 H''(\mu)$: since H is concave, $H'' < 0$, so this term is positive.

Both terms measure information: how much uncertainty H is being destroyed per unit time.

The **HJB equation** balances discounting, stopping, and continuation:

$$\rho\,V(\mu) \;=\; \max\Big\{\rho\,F(\mu),\;\; \sup_{p,\nu,\sigma}\;\big[\mathcal{A}V(\mu) - C(I)\big]\Big\}. \tag{5}$$

Reading the HJB (5)

The left-hand side, $\rho\,V(\mu)$, is the *flow cost of holding the asset* V: at each instant, the DM "loses" $\rho\,V\,dt$ in present value from discounting.
The right-hand side is the maximum of two options:

1. **Stop**: get $\rho F(\mu)$. (The equation says $V(\mu) = F(\mu)$ at stopping beliefs.)
2. **Continue**: the flow gain from learning is $\mathcal{A}V$ (the expected increase in V from learning) minus $C(I)$ (the cost of learning at rate I).

At the optimum, the flow gain from continuing exactly offsets the flow loss from discounting: the DM is indifferent at the margin between stopping and waiting one more instant.

2.3 The main characterisation theorem (binary state)

We now restrict to $|X| = 2$ (binary state), so that $\mu \in [0, 1]$ is a scalar. Define the **experimentation region** $E = \{\mu : V(\mu) > F(\mu)\}$—beliefs at which the DM strictly prefers to continue learning.

Theorem 2 (Theorem 2 of Zhong (2022))**.** *There exists a value function $V \in C^1[0, 1]$ solving (5). In the experimentation region E, the optimal policy is a* ***pure Poisson signal*** *($\sigma \equiv 0$), characterised by five properties:*

(i) ***Poisson dominates Gaussian.*** *At every belief $\mu \in E$ (except possibly at one point μ^*), using a Poisson signal is strictly better than using a Gaussian signal.*

(ii) ***Confirmatory direction.*** *There exists $\mu^* \in \arg\min V$ such that:*

$$\mu > \mu^* \implies \nu(\mu) > \mu \qquad \textit{and} \qquad \mu < \mu^* \implies \nu(\mu) < \mu.$$

The breakthrough pushes the belief further away from *μ^*—it confirms the state the DM already considers most likely.*

(iii) ***Increasing precision.*** *The distance from the jump target to μ^* (a measure of "how decisive" the signal is) increases as μ moves* toward *μ^*. Equivalently, as the DM becomes* less certain*, she demands* more precise *evidence.*

(iv) ***Decreasing frequency.*** *The flow information $I(\mu)$ is increasing in $V(\mu)$. Since V falls as μ approaches μ^*, the DM learns* less intensely *when she is less certain. In particular, the Poisson arrival rate $p(\mu)$ decreases.*

(v) ***Immediate stopping.*** *The jump target $\nu(\mu)$ is always in the stopping region: $\nu(\mu) \in E^C$. The DM stops and acts* immediately *upon receiving the signal.*

2.4 The dynamics in words

Putting the five properties together, the optimal learning dynamics unfold as follows.

Suppose the DM starts at prior $\mu_0 > \mu^*$, so she believes state 1 is more likely.

1. She designs an experiment that, if it "succeeds," makes her *even more* confident in state 1 (confirmatory, property (ii)). The success is rare—it occurs with Poisson intensity $p(\mu_0)$.
2. If the experiment succeeds (the Poisson signal arrives), her belief jumps to $\nu(\mu_0)$ in the stopping region. She immediately chooses the action best suited for state 1 (property (v)).
3. If the experiment does not succeed, she becomes *slightly less confident* in state 1. This is Bayesian updating from the absence of good news: the belief drifts toward μ^*.
4. As the belief drifts toward μ^*:
   - The next experiment is designed with a *higher* passing threshold—it becomes harder to confirm state 1 (property (iii)).
   - Consequently, success becomes *less frequent* (property (iv)).
   - The DM spends less on learning overall.

5. If the belief reaches μ^* before any breakthrough arrives, the DM enters a *stationary phase*: she now seeks two symmetric signals (one confirming each state) in a balanced way. Her belief stays at μ^* until one of the two signals arrives.

R&D interpretation

A firm is deciding which of two technologies to adopt. It currently thinks technology A is better. The optimal R&D strategy is to *test technology A with a very demanding test.* If the test succeeds (rare), adopt technology A immediately. If the test fails (or rather, keeps not succeeding), the firm gradually loses confidence in A. Over time, the test becomes even more demanding but less frequent, and the firm spends less on R&D. Eventually, the firm may become so unsure that it starts testing both technologies symmetrically.

3 Part 3: Why? The Key Tradeoff (8 min)

3.1 The precision–frequency tradeoff

The central economic insight is a tradeoff between two uses of the learning budget: **precision** (how decisive each signal is) and **frequency** (how often a signal arrives).

To isolate this tradeoff, fix the total flow information I (the "budget") and the direction (jump toward $\nu > \mu$). The Poisson signal is described by two parameters: the rate p and the information content per jump $D_H(\nu\|\mu) = H(\mu) - H(\nu) + H'(\mu)(\nu - \mu)$. These are linked by:

$$I = p \cdot D_H(\nu\|\mu).$$

So for a fixed budget I, the DM can choose between *frequent but small jumps* (high p, low D_H—low precision) and *rare but large jumps* (low p, high D_H—high precision). **The question is: where on this frontier does the DM want to be?**

3.2 The gross value function and concavification

The answer comes from a geometric construction. Define the **gross value function**:

$$G(\mu) \;=\; V(\mu) + \lambda\, H(\mu), \qquad \text{where} \qquad \lambda = C'(I^*).$$

Reading G

G combines the value function V (convex) and the uncertainty measure H (concave), weighted by the *shadow price of information* λ. This shadow price is the marginal cost of the optimally chosen information level.

Sannikov & Zhong (2024) give an equivalent interpretation: λ is the price at which the DM can "buy" or "sell" information in a fictitious market. When λ is high, information is expensive, and G tilts toward H (more concave). When λ is low, G tilts toward V (more convex).

Substituting into the optimality condition for ν, one can show that the optimal posterior ν^* satisfies a **concavification condition**:

$$G(\nu) \;\le\; G(\mu) + G'(\mu)(\nu - \mu) \quad \forall\, \nu, \qquad \text{with equality at } \nu = \nu^*. \tag{6}$$

Reading the concavification condition (6)

$G(\mu) + G'(\mu)(\nu - \mu)$ is the tangent line of G at μ. Condition (6) says: G lies *everywhere below* its own tangent line at μ, except at the optimal posterior ν^* where G *touches* the tangent line.

This is precisely the concavification logic from Bayesian persuasion (Kamenica & Gentzkow, 2011). The difference is:

- In Bayesian persuasion, you concavify the sender's payoff to find the optimal *posteriors*. The prior is fixed and the posteriors are the endpoints of the concavified region.
- Here, you concavify G and the *prior* μ is itself one endpoint of the concavified region. The other endpoint is the optimal posterior ν^*.

So the prior is always on the boundary of a concavified region—it is never in the interior.

3.3 How the tradeoff resolves: varying λ

The parameter λ adjusts endogenously with the continuation value. As $V(\mu)$ changes, so does λ, and so does the shape of G, which determines ν^*.

When $V(\mu)$ is high (the DM is quite certain, far from μ^*):

- Discounting costs are high—the DM loses a lot by waiting.
- She wants frequent signals to act quickly.
- λ is high $\Rightarrow G = V + \lambda H$ is *more concave* $\Rightarrow$ the concavified region is *narrow* $\Rightarrow$ the jump $|\nu^* - \mu|$ is *small*.
- Result: **frequent, imprecise** signals.

When $V(\mu)$ is low (the DM is uncertain, near μ^*):

- Discounting costs are low—less to lose from waiting.
- She can afford to wait for rare, conclusive evidence.
- λ is low $\Rightarrow G$ is *less concave* $\Rightarrow$ the concavified region is *wide* $\Rightarrow$ the jump $|\nu^* - \mu|$ is *large*.
- Result: **rare, precise** signals.

This explains properties (iii) and (iv) of Theorem 2: precision increases and frequency decreases as the DM becomes more uncertain.

3.4 Why confirmatory? (Property (ii))

Consider a belief $\mu > \mu^*$ inside the experimentation region. The tangent line of G at μ can only "touch down" on G at some $\nu > \mu$ (on the far side from μ^*). Why?

Within the concavified region $[\mu, \nu^*]$, the value V (and hence λ) is higher than at μ. This follows from the tangent-line condition: any interior point μ' between μ and ν^* must have $G(\mu')$ strictly below the tangent line at μ. Since G at μ' is computed with a *higher* λ (because $V(\mu') > V(\mu)$, giving $\lambda(\mu') > \lambda(\mu)$), the only way this is consistent is if $V(\mu') > V(\mu)$.

Since V is quasi-convex with minimum at μ^*, the direction of increasing V from $\mu > \mu^*$ is *away from* μ^*, i.e., toward $\nu > \mu$. The jump is confirmatory.

Intuition via the no-signal drift

Absent a signal, the compensating drift pushes μ toward μ^* (toward more uncertainty). This makes future signals need to be more precise (larger jump) to reach the stopping region. This increasing-precision path is only consistent with a *decreasing* continuation value—which means the drift goes toward lower V, i.e., toward μ^*. So the jump (which goes in the opposite direction) must point away from μ^*: confirmatory.

3.5 Why no repeated jumps? (Property (v))

Suppose the DM jumps from μ to some ν and then plans to continue learning and eventually jump from ν to ν'. Could this two-step strategy be optimal?

No. The argument is:

1. Since $V(\nu) > V(\mu)$ (the jump is confirmatory toward higher V), the shadow price at ν satisfies $\lambda(\nu) > \lambda(\mu)$.
2. $G_\nu \equiv V + \lambda(\nu)H$ is more concave than $G_\mu \equiv V + \lambda(\mu)H$ (because adding more λH makes G more concave).
3. The concavification condition at ν says $G_\nu(\nu') = G_\nu(\nu) + G'_\nu(\nu)(\nu' - \nu)$. But since G_μ is less concave than G_ν, we have $G_\mu(\nu') > G_\mu(\mu) + G'_\mu(\mu)(\nu' - \mu)$.

4. This implies that jumping *directly* from μ to ν' (skipping ν) achieves a higher benefit-to-information ratio than the current multiplier $\lambda(\mu)$.

So a direct jump strictly dominates the two-step strategy. By induction, the breakthrough happens exactly once: the DM jumps directly to the stopping region.

3.6 Why not Gaussian? (Property (i))

A Gaussian signal corresponds to the *limit* of a Poisson signal where:

$$\text{precision} \to 0, \qquad \text{frequency} \to \infty, \qquad \text{with the product remaining finite.}$$

The first-order condition for an optimal Gaussian signal gives:

$$V''(\mu) + \lambda\, H''(\mu) = 0 \quad \Longleftrightarrow \quad G''(\mu) = 0.$$

That is, the Gaussian signal is optimal *only at beliefs where G has zero curvature*—an inflection point. Since G generically has non-zero curvature almost everywhere, the Gaussian is generically suboptimal.

Economically: the Gaussian is the extreme corner where the DM sacrifices *all* precision for maximum frequency. This is only worthwhile at the boundary of the experimentation region, where the DM is nearly indifferent between stopping and continuing. A slightly more patient DM (strictly inside the experimentation region) always prefers to accept some waiting in exchange for a more informative signal.

3.7 Where Zhong (2022) sits in the bigger picture

Sannikov & Zhong (2024) classify optimal exploration strategies by the DM's time preference. Recall that the stopping payoff takes the form $U(\mu, t) = e^{-\rho t} F(\mu)$. The key property is the **curvature of U in t**:

Time preference	**Curvature of U in t**	**Optimal strategy**
Exponential discounting	Convex (time-risk loving)	Poisson: *Zhong (2022)*
Flow waiting cost	Linear (time-risk neutral)	Any process: *Hébert–Woodford (2023)*
Exogenous deadline	Concave (time-risk averse)	Pure exploration, then burst

Why time-risk attitude matters

The DM's strategy pins down a distribution over stopping times.

A **time-risk loving** DM (exponential discounting) prefers dispersed stopping times: "either stop very quickly or accept a long delay." Poisson learning maximises this dispersion.

A **time-risk neutral** DM (flow waiting cost) cares only about the expected stopping time. Path independence (Theorem 3 of Zhong) then implies that all exploration strategies achieving the same total information are payoff-equivalent. The dynamic choice of "what to learn" is irrelevant.

A **time-risk averse** DM (deadline) prefers concentrated stopping times. She first accumulates information without acting ("pure exploration"), then stops within a narrow window.

The message

The Poisson structure is not an artifact of a specific model. It is the **general consequence of convex time preferences** (which include exponential discounting as a special case) combined with **flexible information acquisition**. Changing the time preference changes the optimal exploration strategy in predictable ways.